MA23313

DISCRETE MATHEMATICS FOR AI

B.Tech Artificial Intelligence & Machine Learning

Semester III

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UNIT I : Mathematical Logic

Propositional Calculus

- Propositions and Connectives
- Syntax and Semantics
- Truth Tables
- Validity and Satisfiability
- Tautology

Connectives and Equivalence

- Functionally Complete Set
- Equivalence
- Logical Identities
- Normal Forms
- Formal Reducibility

Predicate Logic

- Predicates
- Quantifiers
- Nested Quantifiers

Advanced Logic

- Rules of Inference
- Temporal Logic
- Three-Valued Logic

1 Propositions and Connectives

Definition

Logical operators or connectives are symbols used to combine one or more propositions to form compound propositions.

The commonly used logical connectives are:

1. AND ($\wedge$)
2. OR ($\vee$)
3. NOT ($\neg P,\ \sim P,\ P',\ \overline{P}$)
4. If–Then ($\Rightarrow$)
5. If and Only If ($\Leftrightarrow$)
6. Exclusive OR (XOR)
7. NAND
8. NOR
9. XNOR

Definition

Conjunction: When p and q are any two propositions, the conjunction of p and q, denoted by $p \wedge q$, is the compound proposition which is true when both p and q are true and false otherwise.

Truth Table

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Definition

Disjunction: When p and q are any two propositions, the disjunction of p and q, denoted by $p \vee q$, is false only when both p and q are false and true otherwise.

Truth Table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Definition

Negation: The negation of a proposition p is another proposition having the opposite truth value of p. It is denoted by $\neg p,\ \sim p,\ p',\ \overline{p}$.

Truth Table

p	$\neg p$
T	F
F	T

Definition

Conditional (if then $(p \Rightarrow q)$): If p and q are propositions, then the compound proposition "If p then q", denoted by $p \Rightarrow q$, is called a conditional proposition.

Result

The implication $p \Rightarrow q$ is false only when p is true and q is false.

Truth Table

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example

If you come here, then you will get a pass mark.
Let

$$p: \text{ You come here}$$

$$q: \text{ You will get a pass mark}$$

Then the statement is represented by

$$p \Rightarrow q$$

Definition

BiConditional (iff $(p \Leftrightarrow q)$): When p and q are any two propositions, the compound proposition "p if and only if q", denoted by $p \Leftrightarrow q$, is called a **biconditional proposition**, which is true when p and q have the same truth values and false otherwise.

Truth Table

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Definition

The **NAND** operation is the negation of conjunction. It is true when at least one of p or q is false, and false only when both p and q are true.

Truth Table

p	q	$\neg(p \wedge q)$
T	T	F
T	F	T
F	T	T
F	F	T

Definition

Exclusive OR (XOR) is true when exactly one of p and q is true, and false otherwise. It is denoted by $p \oplus q$.

Truth Table

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Definition

The **NOR** operation is the negation of disjunction. It is true only when both p and q are false.

Truth Table

p	q	$\neg(p \vee q)$
T	T	F
T	F	F
F	T	F
F	F	T

Definition

Exclusive NOR (XNOR) is the negation of XOR. It is true when both p and q have the same truth value.

Truth Table

p	q	$\overline{p \oplus q}$
T	T	T
T	F	F
F	T	F
F	F	T

Result

Precedence of Logical Operators (highest to lowest)

Symbol	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\Rightarrow$	4
$\Leftrightarrow$	5

Definition

Tautology: A statement formula that is always true is called a **tautology**.

Example: $p \vee \neg p$ is a tautology.

Truth Table

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Definition

Contradiction: A statement formula that is always false is called a **contradiction**.

Truth Table

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Definition

Contingency: A proposition which is neither a tautology nor a contradiction is called a **contingency**.
Example: $(p \Rightarrow q) \vee (p \wedge q)$ is a contingency.

Truth Table

p	q	$p \Rightarrow q$	$p \wedge q$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	F

Result

Hence $p \wedge q$, $p \vee q$, $p \Rightarrow q$, $p \Leftrightarrow q$ are all contingencies.

Definition

Logically Equivalent: Let A and B be two compound propositions. A and B are said to be **logically equivalent** if A and B have the same set of truth values. Equivalently, A and B are logically equivalent iff $A \Leftrightarrow B$ is a tautology.
Notation: $\Leftrightarrow$, $\equiv$.

Definition

Tautologically Imply: Let A and B be two compound propositions. A is said to **tautologically imply** B iff $A \Rightarrow B$ is a tautology.

Example

Example: $p \Rightarrow (p \vee q)$

Truth Table

p	q	$p \vee q$	$p \Rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

Result

Hence $p \Rightarrow (p \vee q)$ is a tautology.

$$\therefore\ p \Rightarrow (p \vee q)$$

Definition

Principle of Duality: The dual of a compound proposition is the proposition obtained by replacing each $\wedge$ by $\vee$, each $\vee$ by $\wedge$, each T by F and each F by T. The dual of a proposition A is denoted by A^*.

Example

Example:

$$A = (P \vee F) \wedge (Q \vee F)$$
$$A^* = (P \wedge T) \vee (Q \wedge T)$$

Definition

Converse, Inverse, Contrapositive: If $P \to Q$, then:

- The **Converse** is $Q \to P$
- The **Inverse** is $\neg P \to \neg Q$
- The **Contrapositive** is $\neg Q \to \neg P$

Example

Example:

P : It rains, $\quad Q$: I will get wet

$P \to Q$: If it rains, then I will get wet

$Q \to P$: If I get wet, then it will rain

$\neg P \to \neg Q$: If it does not rain, then I will not get wet

$\neg Q \to \neg P$: If I don't get wet, then it will not rain

Result

Equations Involving Biconditionals:

$$(i)\ P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)$$
$$(ii)\ P \leftrightarrow Q \equiv (P \wedge Q) \vee (\neg P \wedge \neg Q)$$
$$(iii)\ P \leftrightarrow Q \equiv \neg(P \oplus Q)$$
$$(iv)\ P \oplus Q \equiv \neg(P \leftrightarrow Q)$$

Result

Laws of Algebra of Propositions:

- **Idempotent Law:**
$$P \vee P \equiv P, \quad P \wedge P \equiv P$$
- **Identity Law:**
$$P \vee F \equiv P, \quad P \wedge T \equiv P$$
- **Domination Law:**
$$P \vee T \equiv T, \quad P \wedge F \equiv F$$
- **Complement Law:**
$$P \vee \neg P \equiv T, \quad P \wedge \neg P \equiv F$$
- **Commutative Law:**
$$P \vee Q \equiv Q \vee P, \quad P \wedge Q \equiv Q \wedge P$$
- **Associative Law:**
$$(P \vee Q) \vee R \equiv P \vee (Q \vee R)$$
$$(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$$
- **Distributive Law:**
$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$
$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$
- **Absorption Law:**
$$P \vee (P \wedge Q) \equiv P, \quad P \wedge (P \vee Q) \equiv P$$
- **De Morgan's Law:**
$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$
$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$
- **Double Negation:**
$$\neg(\neg P) \equiv P$$

Result

Important Implications:

1. $P \wedge Q \Rightarrow P$ (Simplification)
2. $P \wedge Q \Rightarrow Q$ (Simplification)
3. $P \Rightarrow P \vee Q$ (Addition)
4. $Q \Rightarrow P \vee Q$ (Addition)
5. $Q \Rightarrow P \rightarrow Q$
6. $\neg P \Rightarrow P \rightarrow Q$
7. $\neg(P \rightarrow Q) \Rightarrow P$
8. $\neg(P \rightarrow Q) \Rightarrow \neg Q$
9. $P \wedge (P \rightarrow Q) \Rightarrow Q$ (Modus Ponens)
10. $\neg Q \wedge (P \rightarrow Q) \Rightarrow \neg P$ (Modus Tollens)
11. $P \wedge (P \vee Q) \equiv P$ (Disjunctive Syllogism)
12. $(P \vee Q) \wedge \neg Q \equiv P$ (Disjunctive Syllogism)
13. $(P \rightarrow Q) \wedge (Q \rightarrow r) \Rightarrow (P \rightarrow r)$ (Hypothetical Syllogism)
14. $(P \vee Q \rightarrow r) \wedge (Q \rightarrow r) \Rightarrow r$ (Dilemma)
15. $(P, Q) \Rightarrow P \wedge Q$ (Conjunction)
16. $(P \vee Q), (\neg P \vee r) \Rightarrow Q \vee r$ (Resolution)

Result

Equivalence using Conditionals:

1. $p \to q \equiv \neg p \vee q$
2. $p \to q \equiv \neg q \to \neg p$
3. $(p \to q) \to r \equiv (p \wedge \neg q) \to r$
4. $p \vee q \equiv \neg p \to q$
5. $p \wedge q \equiv \neg(p \to \neg q)$
6. $\neg(p \to q) \equiv p \wedge \neg q$
7. $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$
8. $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$
9. $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$
10. $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

Equivalences using Conditionals with Truth Tables

Truth Table

(i) $p \to q \equiv \neg p \vee q$

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Truth Table

(ii) $p \to q \equiv \neg q \to \neg p$

p	q	$p \to q$	$\neg q$	$\neg p$	$\neg q \to \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Truth Table

(iii) $p \to (q \to r) \equiv (p \wedge q) \to r$

p	q	r	$q \to r$	$p \to (q \to r)$	$p \wedge q$	$(p \wedge q) \to r$
T	T	T	T	T	T	T
T	T	F	F	F	T	F
T	F	T	T	T	F	T
T	F	F	T	T	F	T
F	T	T	T	T	F	T
F	T	F	F	T	F	T
F	F	T	T	T	F	T
F	F	F	T	T	F	T

Truth Table

(iv) $p \vee q \equiv \neg p \to q$

p	q	$p \vee q$	$\neg p$	$\neg p \to q$
T	T	T	F	T
T	F	T	F	F
F	T	T	T	T
F	F	F	T	F

Truth Table

(v) $p \wedge q \equiv \neg(p \to \neg q)$

p	q	$\neg q$	$p \to \neg q$	$\neg(p \to \neg q)$	$p \wedge q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	T	F	F

Truth Table

(vi) $\neg(p \to q) \equiv p \wedge \neg q$

p	q	$p \to q$	$\neg(p \to q)$	$\neg q$	$p \wedge \neg q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

Truth Table

(vii) $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$

p	q	r	$p \to q$	$p \to r$	$(p \to q) \wedge (p \to r)$	$q \wedge r$	$p \to (q \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	F	F	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	F	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

Truth Table

(viii) $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$

p	q	r	$p \to r$	$q \to r$	$(p \to r) \wedge (q \to r)$	$(p \vee q) \to r$
T	T	T	T	T	T	T
T	T	F	F	F	F	F
T	F	T	T	T	T	T
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	T	T	T

Truth Table

(ix) $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$

p	q	r	$p \to q$	$p \to r$	$(p \to q) \vee (p \to r)$	$q \vee r$	$p \to (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	F	T

Truth Table

(x) $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

p	q	r	$p \to r$	$q \to r$	$(p \to r) \vee (q \to r)$	$p \wedge q$	$(p \wedge q) \to r$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	F
T	F	T	T	T	T	F	T
T	F	F	F	T	T	F	T
F	T	T	T	T	T	F	T
F	T	F	T	F	T	F	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

Equivalences using Conditionals – Algebraic Proofs

Basic Laws Used

- **Implication law:** $p \to q \equiv \neg p \vee q$
- **De Morgan's laws:** $\neg(p \wedge q) \equiv \neg p \vee \neg q$, $\neg(p \vee q) \equiv \neg p \wedge \neg q$
- **Double negation:** $\neg\neg p \equiv p$
- **Commutativity, associativity, distributivity, idempotent laws, etc.**

Algebraic Proof

(i) $p \to q \equiv \neg p \vee q$
This is the definition of implication. No further steps needed.

Algebraic Proof

(ii) $p \to q \equiv \neg q \to \neg p$ **(Contrapositive)**

$$\begin{aligned} p \to q &\equiv \neg p \vee q \quad \text{(Implication law)} \\ &\equiv q \vee \neg p \quad \text{(Commutativity)} \\ &\equiv \neg q \to \neg p \quad \text{(Implication law, reverse)} \end{aligned}$$

Algebraic Proof

(iii) $p \to (q \to r) \equiv (p \wedge q) \to r$ **(Exportation)**

$$\begin{aligned} p \to (q \to r) &\equiv p \to (\neg q \vee r) \quad \text{(Implication law)} \\ &\equiv \neg p \vee (\neg q \vee r) \quad \text{(Implication law)} \\ &\equiv (\neg p \vee \neg q) \vee r \quad \text{(Associativity)} \\ &\equiv \neg(p \wedge q) \vee r \quad \text{(De Morgan)} \\ &\equiv (p \wedge q) \to r \quad \text{(Implication law)} \end{aligned}$$

Algebraic Proof

(iv) $p \vee q \equiv \neg p \to q$

$$\begin{aligned} \neg p \to q &\equiv \neg(\neg p) \vee q \quad \text{(Implication law)} \\ &\equiv p \vee q \quad \text{(Double negation)} \end{aligned}$$

Algebraic Proof

(v) $p \wedge q \equiv \neg(p \to \neg q)$

$$\begin{aligned} \neg(p \to \neg q) &\equiv \neg(\neg p \vee \neg q) \quad \text{(Implication law)} \\ &\equiv \neg\neg p \wedge \neg\neg q \quad \text{(De Morgan)} \\ &\equiv p \wedge q \quad \text{(Double negation)} \end{aligned}$$

Algebraic Proof

(vi) $\neg(p \to q) \equiv p \wedge \neg q$

$$\begin{aligned} \neg(p \to q) &\equiv \neg(\neg p \vee q) \quad \text{(Implication law)} \\ &\equiv \neg\neg p \wedge \neg q \quad \text{(De Morgan)} \\ &\equiv p \wedge \neg q \quad \text{(Double negation)} \end{aligned}$$

Algebraic Proof

(vii) $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$

$$\begin{aligned} (p \to q) \wedge (p \to r) &\equiv (\neg p \vee q) \wedge (\neg p \vee r) \quad \text{(Implication law)} \\ &\equiv \neg p \vee (q \wedge r) \quad \text{(Distributivity: } (A \vee B) \wedge (A \vee C) \equiv A \vee (B \wedge C)) \\ &\equiv p \to (q \wedge r) \quad \text{(Implication law)} \end{aligned}$$

Algebraic Proof

(viii) $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$

$$\begin{aligned}(p \to r) \wedge (q \to r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) \quad \text{(Implication law)}\\ &\equiv (\neg p \wedge \neg q) \vee r \quad \text{(Distributivity: } (A \vee C) \wedge (B \vee C) \equiv (A \wedge B) \vee C)\\ &\equiv \neg(p \vee q) \vee r \quad \text{(De Morgan)}\\ &\equiv (p \vee q) \to r \quad \text{(Implication law)}\end{aligned}$$

Algebraic Proof

(ix) $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$

$$\begin{aligned}(p \to q) \vee (p \to r) &\equiv (\neg p \vee q) \vee (\neg p \vee r) \quad \text{(Implication law)}\\ &\equiv (\neg p \vee \neg p) \vee (q \vee r) \quad \text{(Associativity, commutativity)}\\ &\equiv \neg p \vee (q \vee r) \quad \text{(Idempotent law)}\\ &\equiv p \to (q \vee r) \quad \text{(Implication law)}\end{aligned}$$

Algebraic Proof

(x) $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

$$\begin{aligned}(p \to r) \vee (q \to r) &\equiv (\neg p \vee r) \vee (\neg q \vee r) \quad \text{(Implication law)}\\ &\equiv (\neg p \vee \neg q) \vee (r \vee r) \quad \text{(Associativity, commutativity)}\\ &\equiv \neg(p \wedge q) \vee r \quad \text{(De Morgan + idempotent } r \vee r \equiv r)\\ &\equiv (p \wedge q) \to r \quad \text{(Implication law)}\end{aligned}$$

NORMAL FORMS

As it was mentioned earlier, the construction of truth tables may not be practical, even with the aid of computer. We therefore consider other procedures known as reduction to normal forms.

If we write the given statement formula in a particular form (interms of $\wedge$, $\vee$ and $\neg$) then it is called *Normal Form*. Here given statement formula and its normal form are equivalent.

It will be convenient to use the word *'Product'* in place of 'conjunction' and 'Sum' in place of 'disjunction' in our current discussion.

Definition

Elementary Product :

A product of the statement variables and their negations in a formula is called *Elementary Product*.

For example, let P and Q be any two atomic variables. Then possible elementry products are

$$P,\ Q,\ \neg P \wedge Q,\ \neg Q \wedge P,\ P \wedge \neg P,\ Q \wedge \neg Q,\ P \wedge \neg P \wedge Q$$

Remark

A statement variable alone is an elementry product. Because P can be written as $P \wedge P$.

Elementary Sum :

Definition

A sum of the two statement variables and their negation is called Elementary Sum.

Let P and Q be any two atomic variables. Then

$$P,\ Q,\ P \vee Q,\ \neg P \vee Q,\ P \vee \neg Q,\ P \vee \neg P \vee Q$$

are some examples of elementry sum.

Disjunctive Normal Forms (DNF) :

Definition

A statement formula which is equivalent to a given formula and which consists of a sum of elementry products is called a *Disjunctive Normal Form* of the given formula,

DNF = (Elementry Product) $\vee$ (Elementry Product) $\vee \cdots \vee$ (Elementry Product)

Conjunctive Normal Forms (CNF) :

For a given statement formula an equivalent formula consisting product of elementry sum is called conjunctive normal form.

CNF = (Elementry Sum) $\wedge$ (Elementry Sum) $\wedge \cdots \wedge$ (Elementry Sum)

Remark

1. Note that DNF and CNF of given statement formula need not be unique.
2. In DNF and CNF, the number of variables in each term need not be same.

Example

Example : 1
Obtain a disjunctive Normal form of $P \wedge (P \rightarrow Q)$

Algebraic Proof

$P \wedge (P \rightarrow Q)$	
$\Rightarrow P \wedge (\neg P \vee Q)$	$(\because\ P \rightarrow Q \Rightarrow \neg P \vee Q)$
$\Rightarrow (P \wedge \neg P) \vee (P \wedge Q)$	(Distributive Rule)

Since the given statement formula is written interms of sum of elementry product.

$\therefore$ DNF of $P \wedge (P \rightarrow Q)$ is $(P \wedge \neg P) \vee (P \wedge Q)$

Example

Example : 2
Obtain DNF of $\neg(P \vee Q) \Leftrightarrow (P \wedge Q)$

Algebraic Proof

$\neg(P \vee Q) \Leftrightarrow (P \wedge Q)$	
$\Rightarrow [\neg(P \vee Q) \wedge (P \wedge Q)] \vee [(P \vee Q) \wedge \neg(P \wedge Q)]$	$\because\ R \Leftrightarrow S \equiv (R \wedge S) \vee (\neg R \wedge \neg S)$
$\Rightarrow (\neg P \wedge \neg Q \wedge P \wedge Q) \vee [(P \vee Q) \wedge (\neg P \vee \neg Q)]$	(DeMorgan's & Associative)
$\Rightarrow (\neg P \wedge \neg Q \wedge P \wedge Q) \vee (P \wedge (\neg P \vee \neg Q)) \vee (Q \wedge (\neg P \vee \neg Q))$	(Distributive Rule)
$\Rightarrow (P \wedge Q \wedge \neg P \wedge \neg Q) \vee (P \wedge \neg P) \vee (P \wedge \neg Q) \vee (Q \wedge \neg P) \vee (Q \wedge \neg Q)$	(Distributive Rule)

$\therefore$ DNF of $\neg(P \vee Q) \Leftrightarrow (P \wedge Q)$ is

$$(P \wedge Q \wedge \neg P \wedge \neg Q) \vee (P \wedge \neg P) \vee (P \wedge \neg Q) \vee (Q \wedge \neg P) \vee (Q \wedge \neg Q).$$

Hence, which is the required DNF.

Example

Example : 3
Obtain DNF of

$$P \to [(P \to Q) \wedge \neg(\neg Q \vee \neg P)]$$

Algebraic Proof

$P \to [(P \to Q) \wedge \neg(\neg Q \vee \neg P)]$	
$\Rightarrow \neg P \vee [(\neg P \vee Q) \wedge \neg(\neg Q \vee \neg P)]$	$(\because P \to Q \equiv \neg P \vee Q)$
$\Rightarrow \neg P \vee [(\neg P \vee Q) \wedge (Q \wedge P)]$	(DeMorgan's Law)
$\Rightarrow \neg P \vee [(\neg P \wedge Q \wedge P) \vee (Q \wedge Q \wedge P)]$	(Distributive Rule)
$\Rightarrow \neg P \vee [(F \wedge Q) \vee (P \wedge Q)]$	$(P \wedge \neg P \Rightarrow F,$ Idempotent)
$\Rightarrow \neg P \vee [F \vee (P \wedge Q)]$	$(P \wedge F = F)$
$\Rightarrow \neg P \vee (P \wedge Q)$	$(P \vee F = P)$

which is the required DNF.

Example

Example : 4
Obtain a disjunctive normal form of

$$(P \wedge \neg(Q \vee R)) \vee (((P \wedge Q) \vee \neg R) \wedge P)$$

Algebraic Proof

$(P \wedge \neg(Q \vee R)) \vee (((P \wedge Q) \vee \neg R) \wedge P)$	
$\Rightarrow (P \wedge (\neg Q \wedge \neg R)) \vee (((P \wedge Q) \vee \neg R) \wedge P)$	$(\because$ DeMorgan's Law)
$\Rightarrow (P \wedge \neg Q \wedge \neg R) \vee ((P \wedge Q) \wedge P) \vee (\neg R \wedge P)$	(Distributive Rule)
$\Rightarrow (P \wedge \neg Q \wedge \neg R) \vee (P \wedge Q) \vee (P \wedge \neg R)$	(Commutative Rule & Idempotent)

which is the required DNF.

PRINCIPAL NORMAL FORMS

Minterms :

Definition

Let P and Q be 2 statement variables. Let us construct all possible formulas which consist of conjunction of P or its negation and conjunction of Q or its negation. None of the formulas should contain both a variable and its negation. Using commutative law if any two terms are equivalent, choose any one of the term. Collect the remaining terms. They are called *minterms*.

For example, let P & Q be two variables, then the minterms are

$$P \wedge Q,\ P \wedge \neg Q,\ \neg P \wedge Q \text{ and } \neg P \wedge \neg Q$$

Remark

Remark – 1 :
If there are 'n' variables then the number of minterms is 2^n.

Remark – 2 :
In elementry product a variable and its negation exist. But in minterm such things does not exist.

Remark – 3 :
Let P, Q and R be 3 variables. Then possible minterms are

(1) $P \wedge Q \wedge R$ (2) $P \wedge Q \wedge \neg R$ (3) $P \wedge \neg Q \wedge R$

(4) $\neg P \wedge Q \wedge R$ (5) $\neg P \wedge \neg Q \wedge R$ (6) $\neg P \wedge Q \wedge \neg R$

(7) $P \wedge \neg Q \wedge \neg R$ and (8) $\neg P \wedge \neg Q \wedge \neg R$

Maxterms :

Definition

Let P and Q be 2 statement variables. Let us construct all possible combination of disjunction P or its negation and Q or its negation. None of the formula should contain both a variable and its negation. Using commutative law, if any two terms are equivalent choose any one of them. Collect the remaining term. They are called *maxterms*.

The possible maxterms with 2 variables are

$$P \vee Q,\ P \vee \neg Q,\ \neg P \vee Q,\ \neg P \vee \neg Q$$

Principal Disjunctive Normal Forms (PDNF) :

Definition

For a given statement formula, an equivalent formula consisting of disjunction of minterms only is known as its *Principal disjunctive normal form* (PDNF).

i.e.,

$$\text{PDNF} = (\text{Minterms}) \vee (\text{Minterms}) \vee \cdots \vee (\text{Minterms})$$

Principal Conjunctive Normal Forms (PCNF) :

Definition

For a given statement formula, an equivalent formula consisting of conjunction of maxterms only is known as its *Principal conjunctive normal form* (PCNF). i.e.,

$$\text{PCNF} = (\text{Maxterms}) \wedge (\text{Maxterms}) \wedge \cdots \wedge (\text{Maxterms})$$

Remark

Remark – 1 :
PDNF is also called sum – of – products canonical form.
PCNF is also called product – of – sums canonical form.

Remark – 2 :
PDNF and PCNF of a given statement formula need not be unique.

Definition

Working Rule to obtain PDNF :

Step – 1 : By applying various equivalence rules simplify the given statement formula, if possible

(i.e., Rewrite the given statement formula interms of $\vee$, $\wedge$ and $\neg$ alone).

Step – 2 : Apply the fact (Each term) $\wedge T$

$$(\because P \wedge T \Leftrightarrow P, \text{ this is possible})$$

Step – 3 : Instead of T, apply $P \vee \neg P$

$$(\because P \vee \neg P \Leftrightarrow T)$$

Step – 4 : Apply Distributive rules.

Step – 5 : Apply Commutative rules.

Remark

1. PCNF and PDNF of a given formula are unique.
2. If a compound proposition is a contradiction, then its PCNF will contain all possible maxterms.
3. If a compound proposition is a tautology, then its PDNF will contain all possible minterms.
4. Two statement formulas are equivalent if their PDNFs are the same.
5. The number of minterms in a PDNF is the same as the number of 'T's in its truth table.
6. The number of maxterms in a PCNF is the same as the number of 'F's in its truth table.

Example

Example : 1
Obtain principal disjunctive normal forms of

$$(a)\ \neg P \vee Q \qquad\qquad (b)\ P \Leftrightarrow Q$$

Solution :

Algebraic Proof

(a) $\neg P \vee Q$

$\neg P \vee Q$	
$\Rightarrow (\neg P \wedge T) \vee (Q \wedge T)$	$(P \wedge T \Leftrightarrow P)$
$\Rightarrow (\neg P \wedge (Q \vee \neg Q)) \vee (Q \wedge (P \vee \neg P))$	$(\because T \Leftrightarrow P \vee \neg P)$
$\Rightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge P) \vee (Q \wedge \neg P)$	(Distributive rules)
$\Rightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (P \wedge Q)$	(Commutative & Idempotent Rules)

which is the required PDNF.

Algebraic Proof

(b) $P \Leftrightarrow Q$

$P \Leftrightarrow Q$	
$\Rightarrow (P \to Q) \wedge (Q \to P)$	$(\because R \Leftrightarrow S \Leftrightarrow (R \to S) \wedge (S \to R))$
$\Rightarrow (\neg P \vee Q) \wedge (\neg Q \vee P)$	$(\because P \to Q \Rightarrow \neg P \vee Q)$
$\Rightarrow (\neg P \wedge (\neg Q \vee P)) \vee (Q \wedge (\neg Q \vee P))$	(Distributive law)
$\Rightarrow (\neg P \wedge \neg Q) \vee (\neg P \wedge P) \vee (Q \wedge \neg Q) \vee (Q \wedge P)$	(Distributive law)

which is the required PDNF.

Example

Example : 2
Obtain principal disjunctive normal forms of

$$(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$$

Solution :

Algebraic Proof

$(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$	
$\Rightarrow ((P \wedge Q) \wedge T) \vee ((\neg P \wedge R) \wedge T) \vee ((Q \wedge R) \wedge T)$	$\because\ P \wedge T \Leftrightarrow P$
$\Rightarrow ((P \wedge Q) \wedge (R \vee \neg R)) \vee ((\neg P \wedge R) \wedge (Q \vee \neg Q)) \vee ((Q \wedge R) \wedge (P \vee \neg P))$	$\because\ P \vee \neg P \Leftrightarrow T$
$\Rightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge R \wedge Q) \vee (\neg P \wedge R \wedge \neg Q) \vee (Q \wedge R \wedge P) \vee (Q \wedge R \wedge \neg P)$	(Distributive Rule)
$\Rightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R) \vee (\neg P \wedge \neg Q \wedge R)$	(Commutative Rule & Idempotent Rule)

$\therefore$

PDNF of $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$ is

$$(P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R) \vee (\neg P \wedge \neg Q \wedge R).$$

Hence, which is the required PDNF.

Example

Example : 3
Obtain PDNF of

$$P \vee (\neg P \wedge Q)$$

Solution :

Algebraic Proof

$P \vee (\neg P \wedge Q)$	
$\Rightarrow (P \wedge T) \vee (\neg P \wedge Q)$	$(\because P \wedge T \Leftrightarrow P)$
$\Rightarrow (P \wedge (Q \vee \neg Q)) \vee (\neg P \wedge Q)$	$(\because P \vee \neg P \Leftrightarrow T)$
$\Rightarrow (P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$	(Distributive Law)

which is the required PDNF.

Example

Example : 4
Obtain PDNF of

$$P \rightarrow \Big((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P)\Big)$$

Solution :

Algebraic Proof

$P \to \Big((P \to Q) \wedge \neg(\neg Q \vee \neg P)\Big)$	
$\Rightarrow \neg P \vee \Big((\neg P \vee Q) \wedge (Q \wedge P)\Big)$	$(\because P \to Q \Leftrightarrow \neg P \vee Q)$ (& DeMorgan's Law)
$\Rightarrow \neg P \vee \Big((\neg P \wedge Q \wedge P) \vee (Q \wedge Q \wedge P)\Big)$	(Distributive Law)
$\Rightarrow \neg P \vee \Big((F \wedge Q) \vee (Q \wedge P)\Big)$	$(P \wedge \neg P \Leftrightarrow F)$ (& Idempotent)
$\Rightarrow \neg P \vee (F \vee (Q \wedge P))$	$(P \wedge F \Leftrightarrow F)$
$\Rightarrow \neg P \vee (Q \wedge P)$	$(P \vee F \Leftrightarrow P)$
$\Rightarrow (\neg P \wedge T) \vee (Q \wedge P)$	$(P \wedge T \Leftrightarrow P)$
$\Rightarrow (\neg P \wedge (Q \vee \neg Q)) \vee (Q \wedge P)$	$(T \Leftrightarrow P \vee \neg P)$
$\Rightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge P)$	(Distributive Law)

$$(\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (P \wedge Q)$$

which is the required PDNF.

Definition

Working Rule to obtain PCNF :

Step – 1 : Using Equivalence Rule, simplify the given statement formula, if possible.

(i.e., Rewrite the given statement formula interms of $\vee$, $\wedge$ and $\neg$ alone).

Step – 2 : Apply the fact (Each term)

$$\vee F \qquad (\because P \vee F \Leftrightarrow P)$$

Step – 3 : Instead of F, apply the fact

$$P \wedge \neg P$$

Step – 4 : Apply Distributive rules.

Step – 5 : Apply Commutative rules.

Example

Example : 1
Obtain the principal conjunctive normal form of the formula

$$(\neg P \to R) \wedge (Q \Leftrightarrow P)$$

Solution :

Algebraic Proof

$(\neg P \to R) \wedge ((Q \to P) \wedge (P \to Q))$	$(\because P \Leftrightarrow Q \Leftrightarrow (P \to Q) \wedge (Q \to P))$
$\Rightarrow (P \vee R) \wedge (\neg Q \vee P) \wedge (\neg P \vee Q)$	$(\because P \to Q \Leftrightarrow \neg P \vee Q)$
$\Rightarrow ((P \vee R) \vee F) \wedge ((\neg Q \vee P) \vee F) \wedge ((\neg P \vee Q) \vee F)$	$(\because P \vee F = P)$
$\Rightarrow ((P \vee R) \vee (Q \wedge \neg Q)) \wedge ((\neg Q \vee P) \vee (R \wedge \neg R)) \wedge ((\neg P \vee Q) \vee (R \wedge \neg R))$	$(\because F \Leftrightarrow P \wedge \neg P)$
$\Rightarrow (P \vee R \vee Q) \wedge (P \vee R \vee \neg Q) \wedge (\neg Q \vee P \vee R) \wedge (\neg Q \vee P \vee \neg R) \wedge (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$	(Distributive Law)
$\Rightarrow (P \vee Q \vee R) \wedge (P \vee \neg Q \vee R) \wedge (P \vee \neg Q \vee \neg R) \wedge (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$	(Commutative Law)

which is the required PCNF.

Example

Example : 2
Obtain the PCNF of the following

$$P \to (\neg P \wedge (Q \to P))$$

Solution :

Algebraic Proof

$P \to (\neg P \wedge (Q \to P))$	
$\Leftrightarrow \neg P \vee (\neg P \wedge (\neg Q \vee P))$	$(\because P \to Q \Leftrightarrow \neg P \vee Q)$
$\Leftrightarrow (\neg P \vee \neg P) \wedge (\neg P \vee \neg Q \vee P)$	(Distributive law)
$\Leftrightarrow \neg P \wedge (T \vee \neg Q)$	$(\because$ Idempotent & $P \vee \neg P \Leftrightarrow T)$
$\Leftrightarrow \neg P \wedge T$	$(P \wedge T \Leftrightarrow T)$
$\Leftrightarrow \neg P$	$(P \wedge T \Leftrightarrow P)$
$\Leftrightarrow (\neg P \vee F)$	$(P \vee F \Leftrightarrow P)$
$\Leftrightarrow \neg P \vee (Q \wedge \neg Q)$	$(F = P \wedge \neg P)$
$\Leftrightarrow (\neg P \vee Q) \wedge (\neg P \vee \neg Q)$	(Distributive law)

which is the required PCNF.

Example

Example : 3
Obtain the PCNF of the following

$$\neg(P \Leftrightarrow Q)$$

Solution :

Algebraic Proof

$\neg(P \Leftrightarrow Q)$	
$\Leftrightarrow \neg((P \to Q) \wedge (Q \to P))$	
$\Leftrightarrow \neg((\neg P \vee Q) \wedge (\neg Q \vee P))$	$(P \to Q \Rightarrow \neg P \vee Q)$
$\Leftrightarrow (P \wedge \neg Q) \vee (Q \wedge \neg P)$	(Distributive law)
$\Leftrightarrow ((P \wedge \neg Q) \vee Q) \wedge ((P \wedge \neg Q) \vee \neg P)$	(Distributive law)
$\Leftrightarrow (P \vee Q) \wedge (\neg Q \vee Q) \wedge (P \vee \neg P) \wedge (\neg Q \vee \neg P)$	(Distributive law)
$\Leftrightarrow (P \vee Q) \wedge T \wedge T \wedge (\neg Q \vee \neg P)$	$(P \vee \neg P \Leftrightarrow T)$
$\Leftrightarrow (P \vee Q) \wedge (\neg P \vee \neg Q)$	$(P \wedge T = P)$

which is the required PCNF.

Example

Example : 4
Obtain the PCNF of

$$(Q \to P) \wedge (\neg P \wedge Q)$$

Solution :

Algebraic Proof

$(Q \to P) \wedge (\neg P \wedge Q)$	
$\Rightarrow (\neg Q \vee P) \wedge (\neg P \wedge Q)$	$(P \to Q \Rightarrow \neg P \vee Q)$
$\Rightarrow (\neg Q \vee P) \wedge ((\neg P \wedge Q) \vee F)$	$(P \vee F = P)$
$\Rightarrow (\neg Q \vee P) \wedge ((\neg P \wedge Q) \vee (P \wedge \neg P))$	$(P \wedge \neg P \Rightarrow F)$
$\Rightarrow (\neg Q \vee P) \wedge ((\neg P \wedge Q) \vee P)$ $\wedge((\neg P \wedge Q) \vee \neg P)$	(Distributive law)
$\Rightarrow (\neg Q \vee P) \wedge (\neg P \vee P) \wedge (Q \vee P)$ $\wedge(\neg P \vee \neg P) \vee (Q \vee \neg P)$	(Distributive law)
$\Rightarrow (\neg Q \vee P) \wedge T \wedge (Q \vee P)$ $\wedge \neg P \vee (Q \vee \neg P)$	$(P \vee \neg P \Rightarrow T)$
$\Rightarrow (P \vee \neg Q) \wedge (P \vee Q) \wedge \neg P$ $\vee(Q \vee \neg P)$	$(P \wedge T = P$ & Idempotent law)

which is the required PCNF.

Inference Theory

Result

Rules for inference theory

Rule P : A premise may be introduced at any point in the derivation.

Rule T : A formula S may be introduced in a derivation if S is a tautologically implied by any one or more of the preceding formulas in the derivation.

Rule CP : If we can derive S from R and a set of premises, then we can derive $R \to S$ from the set of premises alone.

Remark

Note 1 : Rule CP is also called the deduction theorem.

Note 2 : Whenever the assumed premise is used in the derivation, then the method of derivation is called indirect method of derivation.

Example

Example 1 :
Demonstrate that R is a valid inference from the premises

$$P \to Q, \quad Q \to R \quad \text{and} \quad P.$$

Algebraic Proof

Solution :

	Step	Derivation	Rule
$\{1\}$	(1)	$P \to Q$	P
$\{2\}$	(2)	P	P
$\{1,2\}$	(3)	Q	T, (1), (2) and I_{11} $[P,\ P \to Q \Rightarrow Q]$
$\{4\}$	(4)	$Q \to R$	P
$\{1,2,4\}$	(5)	R	T, (3), (4) and I_{11} $[Q,\ Q \to R \Rightarrow R]$

Example

Example 2 : Show that $R \vee S$ follows logically from the premises

$$C \vee D, \qquad (C \vee D) \to \neg H, \qquad \neg H \to (A \wedge \neg B)$$

and

$$(A \wedge \neg B) \to (R \vee S).$$

Algebraic Proof

Solution :

$\{1\}$	(1)	$(C \vee D) \to \neg H$	P
$\{2\}$	(2)	$\neg H \to (A \wedge \neg B)$	P
$\{1,2\}$	(3)	$(C \vee D) \to (A \wedge \neg B)$	T, (1), (2) and I_{13}
$\{4\}$	(4)	$(A \wedge \neg B) \to (R \vee S)$	P
$\{1,2,4\}$	(5)	$(C \vee D) \to (R \vee S)$	T, (3), (4) and I_{13}
$\{6\}$	(6)	$C \vee D$	P
$\{1,2,4,6\}$	(7)	$R \vee S$	T, (5), (6) and I_{11}

The two tautologies frequently used in the above derivations are I_{13}, known as hypothetical syllogism, and I_{11}, known as modus ponens.

Example

Example 3 :
Show that $S \vee R$ is tautologically implied by

$$(P \vee Q) \wedge (P \to R) \wedge (Q \to S)$$

Algebraic Proof

Solution :

$\{1\}$	(1)	$P \vee Q$	P
$\{1\}$	(2)	$\neg P \to Q$	T, (1) and E_1, E_{16}
$\{3\}$	(3)	$Q \to S$	P
$\{1,3\}$	(4)	$\neg P \to S$	T, (2), (3) and I_{13}
$\{1,3\}$	(5)	$\neg S \to P$	T, (4), E_{18} and E_1
$\{6\}$	(6)	$P \to R$	P
$\{1,3,6\}$	(7)	$\neg S \to R$	T, (5), (6) and I_{13}
$\{1,3,6\}$	(8)	$S \vee R$	T, (7), E_{16} and E_1

Example

Example 4 :
Show that

$$R \wedge (P \vee Q)$$

is a valid conclusion from the premises

$$P \vee Q, \qquad Q \to R, \qquad P \to M$$

and

$$\neg M.$$

[A.U. N/D 2007]

Algebraic Proof

Solution :

$\{1\}$	(1)	$P \to M$	P
$\{2\}$	(2)	$\neg M$	P
$\{1,2\}$	(3)	$\neg P$	T, (1), (2) and I_{12}
$\{4\}$	(4)	$P \vee Q$	P
$\{1,2,4\}$	(5)	Q	T, (3), (4) and I_{10}
$\{6\}$	(6)	$Q \to R$	P
$\{1,2,4,6\}$	(7)	R	T, (5), (6) and I_{11}
$\{1,2,4,6\}$	(8)	$R \wedge (P \vee Q)$	T, (4), (7) and I_9

Definition

Consistency of Premises and Indirect Method of Proof
A set of formulas

$$H_1,\ H_2,\ \ldots,\ H_m$$

is said to be consistent if their conjunction has the truth value T for some assignment of the truth values to the atomic variables appearing in

$$H_1,\ H_2,\ \ldots,\ H_m.$$

If for every assignment of the truth values to the atomic variables, atleast one of the formulas

$$H_1,\ H_2,\ \ldots,\ H_m$$

is false, so that their conjunction is identically false, then the formulas

$$H_1,\ H_2,\ \ldots,\ H_m$$

are called inconsistent.
In other way, a set of formulas

$$H_1,\ H_2,\ \ldots,\ H_m$$

is inconsistent if their conjunction implies a contradiction, that is

$$H_1 \wedge H_2 \wedge \cdots \wedge H_m \Rightarrow R \wedge \neg R$$

where R is any formula.
Note that

$$R \wedge \neg R$$

is a contradiction, and it is necessary and sufficient for the implication that

$$H_1 \wedge H_2 \wedge \cdots \wedge H_m$$

be a contradiction.

Definition

Indirect method of proof :
The notion of inconsistency is used in a procedure called proof by contradiction or reduction ad absurdum or indirect method of proof.
The technique of indirect method of proof runs as follows :

1. Introduce the negation of the desired conclusion as a new premise.
2. From the new premise, together with the given premises, derive a contradiction.
3. Assert the desired conclusion as a logical inference from the premises.

Example

Example 1 :
Show that

$$\neg(P \wedge Q)$$

follows from

$$\neg P \wedge \neg Q.$$

Algebraic Proof

Solution :
We introduce

$$\neg\neg(P \wedge Q)$$

as an additional premise and show that this additional premise leads to a contradiction.

$\{1\}$	(1)	$\neg\neg(P \wedge Q)$	P (assumed)
$\{1\}$	(2)	$P \wedge Q$	T, (1) and E_1
$\{1\}$	(3)	P	T, (2) and I_1
$\{4\}$	(4)	$\neg P \wedge \neg Q$	P
$\{4\}$	(5)	$\neg P$	T, (4), I_1
$\{1,4\}$	(6)	$P \wedge \neg P$	T, (3), (5), I_9

Example

Example 2 :
Show that the following set of premises is inconsistent.

If the contract is valid, then John is liable for penalty.
If John is liable for penalty, he will go bankrupt.
If the bank will loan him money, he will not go bankrupt.
As a matter of fact, the contract is valid and the bank will loan him money.

Algebraic Proof

Solution :
We indicate the given statements as follows :

V : The contract is valid.

L : John is liable for penalty.

M : Bank will loan him money.

B : He will go bankrupt.

Then the given premises are

$$V \to L, \qquad L \to B, \qquad M \to \neg B, \qquad V \wedge M$$

[1]	(1)	$V \to L$	Rule P
[2]	(2)	$L \to B$	Rule P
[1, 2]	(3)	$V \to B$	Rule T, (1),(2), law of hypothetical syllogism
[4]	(4)	$V \wedge M$	Rule P
[4]	(5)	V	Rule T, (4)
[4]	(6)	M	Rule T, (4)
[1, 2, 4]	(7)	B	Rule T, (3),(5), modus ponens
[8]	(8)	$M \to \neg B$	Rule P
[4, 8]	(9)	$\neg B$	Rule T, (6),(8)
[1, 2, 4, 8]	(10)	$B \wedge \neg B$	Rule T, (7),(9), contradiction

Thus the given set of premises leads to a contradiction and hence it is inconsistent.

Example

Example 3 :
Show that the following premises are inconsistent.

1. If Jack misses many classes through illness, then he fails high school.
2. If Jack fails high school, then he is uneducated.
3. If Jack reads a lot of books, then he is not uneducated.
4. Jack misses many classes through illness and reads a lot of books.

[A.U. A/M, 2004]

Algebraic Proof

Solution :

E : Jack misses many classes.

S : Jack fails high school.

A : Jack reads a lot of books.

H : Jack is uneducated.

The premises are

$$E \to S, \qquad S \to H, \qquad A \to \neg H$$

and

$$E \wedge A.$$

$\{1\}$	(1)	$E \to S$	P
$\{2\}$	(2)	$S \to H$	P
$\{1,2\}$	(3)	$E \to H$	T, (1), (2) and I_{13}
$\{4\}$	(4)	$A \to \neg H$	P
$\{4\}$	(5)	$H \to \neg A$	T, (4), E_{18}
$\{1,2,4\}$	(6)	$E \to \neg A$	T, (3), (5), I_{13}
$\{1,2,4\}$	(7)	$\neg E \vee \neg A$	T, (6), E_{16}
$\{1,2,4\}$	(8)	$\neg(E \wedge A)$	T, (7), E_8
$\{9\}$	(9)	$E \wedge A$	P
$\{1,2,4,9\}$	(10)	$(E \wedge A) \wedge \neg(E \wedge A)$	T, (8), (9), I_9

Example

Example 4 :
Using indirect method of proof, derive

$$P \to \neg S$$

from

$$P \to (Q \vee R), \quad Q \to \neg P, \quad S \to \neg R, \quad P.$$

Algebraic Proof

Solution :
The desired result is

$$P \to \neg S.$$

Its negation is

$$P \wedge S.$$

Since

$$P \wedge S \Leftrightarrow \neg(\neg P \vee \neg S) \Leftrightarrow \neg(P \to \neg S)$$

is a tautology from the law of negation for implication.
We include

$$P \wedge S$$

as an additional premise.

[1]	(1)	$P \to (Q \vee R)$	Rule P
[2]	(2)	P	Rule P
[1, 2]	(3)	$Q \vee R$	Rule T, (1),(2), modus ponens (I_{11})
[4]	(4)	$S \to \neg R$	Rule P
[5]	(5)	$P \wedge S$	Rule P (new premise)
[5]	(6)	S	Rule T, (5)
[4, 5]	(7)	$\neg R$	Rule T, (4),(6), modus ponens (I_{11})
[1, 2, 4, 5]	(8)	Q	Rule T, (3),(7), I_{10}
[9]	(9)	$Q \to \neg P$	Rule P
[1, 2, 4, 5, 9]	(10)	$\neg P$	Rule T, (8),(9), modus ponens
[1, 2, 4, 5, 9]	(11)	$P \wedge \neg P$	Rule T, (2),(10), contradiction

Thus additional premise

$$P \wedge S$$

and the given premises together lead to a contradiction.
So

$$\neg(P \wedge S)$$

is derivable from

$$P \to (Q \vee R),\ Q \to \neg P,\ S \to \neg R,\ P.$$

Example

Example 5 :
Prove by indirect method that

$$(\neg Q), \quad P \to Q, \quad P \vee R \Rightarrow R$$

Algebraic Proof

Solution :
The desired result is R.
Include its negation as a new premise.

[1]	(1)	$P \vee R$	Rule P
[2]	(2)	$\neg R$	Rule P (additional premise)
[1, 2]	(3)	P	Rule T, (1),(2)
[4]	(4)	$P \to Q$	Rule P
[1, 2, 4]	(5)	Q	Rule T, (3),(4), modus ponens
[6]	(6)	$\neg Q$	Rule P
[1, 2, 4, 6]	(7)	$Q \wedge \neg Q$	Rule T, (5),(6), contradiction

The new premise, together with the given premises, leads to a contradiction.
Thus

$$(\neg Q), \quad P \to Q, \quad P \vee R \Rightarrow R.$$

Result

Validity of (verbal arguments)
In every day life, we come across argument expressed in (English) sentences. We can represent the premises in symbols and verify their validity.

Example

Example 1 :
Determine the validity of the following argument.

If two sides of a triangle are equal, then opposite angles are equal.

Two sides of a triangle are not equal.

$\therefore$

The opposite angles are not equal.

Algebraic Proof

Solution :
Let

P : Two sides of a triangle are equal.

Q : The two opposite angles are equal.

The premises can be represented as

$$P \to Q$$

and

$$\neg P$$

and the conclusion as

$$\neg Q.$$

If the argument is a valid one then

$$((P \to Q) \wedge \neg P) \to \neg Q$$

will be a tautology.
Let us now construct the truth table for

$$((P \to Q) \wedge \neg P) \to \neg Q.$$

Truth Table

P	Q	$P \to Q$	$\neg P$	$(P \to Q) \wedge (\neg P)$	$\neg Q$	$((P \to Q) \wedge (\neg P)) \to \neg Q$
T	T	T	F	F	F	T
T	F	F	F	F	T	T
F	T	T	T	T	F	F
F	F	T	T	T	T	T

From the truth table we can infer that

$$((P \to Q) \wedge \neg P) \to \neg Q$$

is not a tautology.Hence the argument is not valid.

Example

Example 2 :
Show the following argument is valid.

"My father praises me only if I can be proud of myself. Either I do well in sports or I cannot be proud of myself. If I study hard, then I cannot do well in sports. Therefore, if father praises me, then I do not study well."

Algebraic Proof

Solution :
Let

$$A : \text{My father praises me}$$

$$B : \text{I can be proud of myself}$$

$$C : \text{I do well in sports}$$

$$D : \text{I study hard}$$

Then the given premises are

$$A \to B, \qquad C \vee \neg B, \qquad D \to \neg C$$

and the conclusion is

$$A \to \neg D.$$

For, let us assume A as one more premise.

Set	Step	Derivation	Rule
$\{1\}$	(1)	A	Rule P (assumed)
$\{2\}$	(2)	$A \to B$	Rule P
$\{1,2\}$	(3)	B	Rule T, (1), (2), I_{11}
$\{4\}$	(4)	$C \vee \neg B$	Rule P
$\{1,2,3,4\}$	(5)	C	Rule T, (3), (4), I_{10}
$\{6\}$	(6)	$D \to \neg C$	Rule P
$\{6\}$	(7)	$C \to \neg D$	Rule T, (5), E_{18}, E_1
$\{5,7\}$	(8)	$\neg D$	Rule T, (5), (7), I_{11}
$\{1,8\}$	(9)	$A \to \neg D$	Rule CP, (1), (8)

Hence it is a valid argument.

Example

Problem 1
Derive using Rule CP.

$$\neg P \vee Q, \quad \neg Q \vee R, \quad R \to S \;\Rightarrow\; P \to S$$

Algebraic Proof

Solution :
To prove $P \to S$, assume P as an additional premise.

Set	Step	Derivation	Rule
$\{1\}$	(1)	$\neg P \vee Q$	Rule P
$\{2\}$	(2)	$\neg Q \vee R$	Rule P
$\{3\}$	(3)	$R \to S$	Rule P
$\{4\}$	(4)	P	Rule P (Assumed)
$\{1,4\}$	(5)	Q	Rule T, (1), (4), I_{10}
$\{1,2,4\}$	(6)	R	Rule T, (2), (5), I_{10}
$\{1,2,3,4\}$	(7)	S	Rule T, (3), (6), I_{11}
$\{4,7\}$	(8)	$P \to S$	Rule CP, (4), (7)

Hence,

$$\neg P \vee Q, \quad \neg Q \vee R, \quad R \to S \Rightarrow P \to S.$$

Example

Problem 2
Using Rule CP. Derive

$$P \to Q \;\Rightarrow\; P \to (P \wedge Q)$$

Algebraic Proof

Solution :
To prove $P \to (P \wedge Q)$, assume P as an additional premise.

Set	Step	Derivation	Rule
$\{1\}$	(1)	$P \to Q$	Rule P
$\{2\}$	(2)	P	Rule P (Assumed)
$\{1,2\}$	(3)	Q	Rule T, (1), (2), I_{11}
$\{1,2\}$	(4)	$P \wedge Q$	Rule T, (2), (3), I_9
$\{2,4\}$	(5)	$P \to (P \wedge Q)$	Rule CP, (2), (4)

Hence,

$$P \to Q \Rightarrow P \to (P \wedge Q).$$

Definition

Temporal Logic

Temporal Logic is a formal system used to represent and reason about propositions that "change" over time.

While classical logic deals with absolute truth, temporal logic deals with when a proposition is true.

It is fundamental in Computer Science for verifying hardware and software systems.

Definition

Three Valued Logic

Classical logic is bivalent, meaning a "statement" must be either True (T) or False (F).

Now Three Valued Logic introduces the third truth value often denoted as

$$U$$

(unknown / undetermined)

to account for ambiguity, missing information and incomplete statements.

Truth Table

Truth Table for Conjunction ($\wedge$)

$\wedge$	T	F	U
T	T	F	U
F	F	F	F
U	U	F	U

Truth Table

Truth Table for Disjunction ($\vee$)

$\vee$	T	F	U
T	T	T	T
F	T	F	U
U	T	U	U

Result

$$\neg T = F$$

$$\neg F = T$$

$$\neg U = U$$

THE PREDICATE CALCULUS

The predicate calculus deals with the study of predicates.

Consider the following statement:

"Ram is a boy"

In the above statement, *"is a boy"* is the predicate and the name *"Ram"* is the subject.

If we denote the predicate "is a boy" by B and subject "Ram" by r, then the statement "Ram is a boy" can be represented as

$$B(r).$$

Remark

- Always we denote predicates by capital letters and the subjects by small letters.
- In general, any statement of the type "S is P" where P is a predicate and S is the subject can be denoted by $P(S)$.

Some More Examples

(1) "x is a man".

Here, Predicate is *"is a man"* and it is denoted by M. Subject is "x" and it is denoted by x.

$\therefore$ The given statement "x is a man" can be denoted by $M(x)$.

(2) "Vaisali is a girl".

Here again, Predicate is *"is a girl"* and it is denoted by G and subject is "Vaisali" and it is denoted by v.

$\therefore$ The given statement "Vaisali is a girl" can be denoted by $G(v)$.

(3) "Sam is poor and Ram is intelligent".

The statement "Sam is poor" can be represented by $P(s)$ where P represents predicate "is poor" and s represents subject "Sam". The statement *"Ram is intelligent"* can be represented by $I(r)$ where I represents predicate *"is intelligent"* and r represents subject *"Ram"*.

$\therefore$ The given statement "Sam is poor and Ram is intelligent"

can be symbolized as

$$P(s) \wedge I(r).$$

Remark

From Example (3), it follows that 2 or more statements can be combined to form compound statements by using the same set of connectives i.e.,

$$\{\wedge, \vee, \rightarrow, \leftrightarrow, \neg\}$$

which was discussed earlier.

1 – Place predicate, 2 – Place predicate, 3 – Place predicate, 4 – Place predicate

Consider the following statements:

1. Pavithra is rich.
2. Renuka is shorter than Manjula.
3. Mr. X sits between Mr. Y and Mr. Z.
4. Green and Miller played bridge against Johnson & Smith.

In statement (1), the predicate *"is rich"* is a *1 – place predicate* because there is only one name of object "Pavithra" associated with a predicate.

Result

If there is only one name of object associated with a predicate then it is known as 1 – place predicate.

In statement (2), the predicate *"is shorter than"* is a *2 – place predicate* because there are two names of objects "Renuka & Manjula" which are associated with a predicate.

Result

If there are 2 names of objects associated with a predicate then it is known as 2 – place predicate.

In statement (3), the predicate *"sits between"* is a *3 – place predicate* because there are 3 names of objects "Mr. X, Mr. Y & Mr. Z" which are associated with a predicate.

Result

If there are 3 names of objects associated with a predicate then it is known as 3 – place predicate.

In statement (4), the predicate *"played bridge against"* is a *4 – place predicate* because there are 4 names of objects "Green, Miller, Johnson & Smith" which are associated with a predicate.

Result

If there are 4 names of objects associated with a predicate then it is known as 4 – place predicate.

Definition

n – place predicate :
In general, an n – place predicate is a predicate requiring n names of objects where $n > 0$. It is denoted by

$$P(a_1, a_2, \ldots, a_n)$$

where $a_1, a_2, \ldots, a_n$ are the names of objects associated with predicate and P is a predicate.

Remark

Remark :

(a) Statement (1) can be symbolized as $R(p)$ where R represents predicate "is rich" and p represents subject "Pavithra".

(b) Statement (2) can also be symbolized as $S(r, m)$ where S represents predicate "is shorter than" and r, m represents subjects "Renuka & Manjula".

(c) Statement (3) can be symbolized as $S(x, y, z)$ where S represents predicate "sits between" and x, y, z represents subjects "Mr. X, Mr. Y & Mr. Z".

(d) In a similar way,

Statement (4) can be symbolized as

$$P(g, m, j, s).$$

THE STATEMENT FUNCTIONS, VARIABLES, QUANTIFIERS

Definition

Statement Functions :
A simple statement function of one variable is defined to be an expression consisting of a predicate symbol and an individual variable.

Example 1 :

$$M(x): \quad x \text{ is mortal} \qquad \cdots (*)$$

Remark

A statement function becomes a statement when the variable is replaced by the name of any object.

Example 2 :
From $(*)$ we have,

$$M(x): \quad x \text{ is mortal} \qquad \cdots (1)$$

which represents statement function.
Replacing x by a name say Sam in (1), we get

$$M(\text{Sam}): \quad \text{Sam is mortal} \qquad \cdots (2)$$

Replacing x by a name say Rose in (1), we get

$$M(\text{Rose}): \quad \text{Rose is mortal} \qquad \cdots (3)$$

Obviously (2) & (3) represents statements.
Thus as the variable x is replaced by the name of any object in (1), the statement function becomes a statement.

Here we call x to be the place holder.

Definition

Compound statement function is obtained by combining one or more simple statement functions using logical connective.

Example :
Let

$$M(x): \quad x \text{ is a man}$$

and

$$H(x): \quad x \text{ is a mortal}$$

be the 2 simple statement functions.
Then we can form compound statement functions as

$$(i) \qquad M(x) \wedge H(x)$$

$$(ii) \qquad M(x) \vee H(x)$$

$$(iii) \qquad M(x) \rightarrow H(x)$$

$$(iv) \qquad \neg H(x)$$

$$(v) \qquad M(x) \leftrightarrow \neg H(x)$$

Definition

A statement function of 2 variables is an expression consisting of a predicate symbol and 2 individual variables.

Example :

$$G(x, y): \quad x \text{ is taller than } y \qquad \cdots (1)$$

which is an example of 2 variable statement function.
In order to obtain a statement, replace x and y by the names of objects.

$$\therefore$$

If we replace x by Laxmanan and y by Karthikeyan in (1) then we get a statement as

$$G(l, k): \quad \text{Laxmanan is taller than Karthikeyan.}$$

QUANTIFIERS :

Quantifier is one which is used to quantify the nature of variables.

There are 2 important quantifiers which are for *"all"* and for *"some"* where *"some"* means *"atleast one"*.

Definition

Universal Quantifier :
The quantifier "for all x" is called the universal quantifier. It is denoted by the symbol "$(\forall x)$" or (x). The universal quantifier is equivalent to each of the following phrases.

1. For all x
2. For every x
3. For each x
4. Everything x is such that
5. Each thing x is such that

Example

Example 1 :

"Every apple is red".

The above statement can be restated as follows

$$\text{For all } x, \text{ if } x \text{ is an apple then } x \text{ is red} \qquad \cdots (*)$$

Now, we will translate it into symbolic form using universal quantifier.
Define

$$A(x): \quad x \text{ is an apple}$$

$$R(x): \quad x \text{ is red}$$

$$\therefore \quad \text{We write } (*) \text{ into symbolic form as}$$

$$(\forall x)(A(x) \to R(x))$$

Remark

The universal quantifier is also denoted by "(x)" or $(\forall x)$.

Example

Example 2 :

"Everything is yellow".

The above statement can be restated as follows

$$\text{For all } x, \ x \text{ is yellow} \qquad \cdots (A)$$

Now, we shall translate it into symbolic form using universal quantifier.
Let

$$Y(x): \quad x \text{ is yellow}$$

$$\therefore \quad \text{We write } (A) \text{ into symbolic form as}$$

$$(\forall x)(Y(x))$$

Definition

Existential Quantifier :
The quantifier for "some x" is called the existential quantifier. It is denoted by the symbol "$(\exists x)$". The existential quantifier is also equivalent to each of the following phrases.

1. For some x
2. Some x such that
3. There exists an x such that
4. There is an x such that
5. There is atleast one x such that

Example

Example 1 :

"Some men are clever".

The above statement can be restated as

"there is an x such that x is a man and x is clever" $\cdots(B)$

We will translate it into symbolic form using Existential quantifier.
Let

$$M(x): \quad x \text{ is a man}$$

and

$$C(x): \quad x \text{ is clever}$$

$\therefore$ We write (B) into symbolic form as

$$(\exists x)(M(x) \wedge C(x))$$

Example

Example 2 :
There is a man.
The above statement can be restated as

$$\text{“there exists an } x \text{ such that } x \text{ is a man”} \qquad \cdots (C)$$

Now, we will translate it into symbolic form using Existential quantifier.
Let

$$M(x): \quad x \text{ is a man}$$

$$\therefore \quad \text{We write } (C) \text{ as } (\exists x)(M(x))$$

Example

Example : 1
Let

$$M(x): \quad x \text{ is a mammal}$$

$$W(x): \quad x \text{ is warm blooded}$$

Translate into formula:

Every mammal is warm blooded.

Algebraic Proof

The given statement “Every mammal is warm blooded” can be restated as “For all x, if x is a mammal then x is warm blooded”.

$$\therefore \quad \text{Its symbolic form is}$$

$$(\forall x)(M(x) \to W(x))$$

Example

Example : 2
Let

$$G(x, y): \quad x \text{ is taller than } y$$

Translate the following into formula:
For any x and any y if x is taller than y then it is not true that y is taller than x.

Algebraic Proof

Given

$$G(x, y): \quad x \text{ is taller than } y$$

$$\therefore \quad G(y, x): \quad y \text{ is taller than } x$$

$$\therefore \quad \text{The symbolic form of the given statement is}$$

$$(\forall x)(\forall y)(G(x, y) \rightarrow \neg G(y, x))$$

Example

Example : 3
Symbolize the following statement:

"All the world loves a lover".

Algebraic Proof

Let

$$P(x): \quad x \text{ is a person}$$

$$L(x): \quad x \text{ is a lover}$$

$$R(x, y): \quad x \text{ loves } y$$

The given statement can be restated as
"For all x if x is a person then for all y, if y is a person and y is a lover then x loves y".

$$\therefore \quad \text{Its symbolic form is}$$

$$(\forall x)\Big(P(x) \rightarrow (\forall y)\big((P(y) \wedge L(y)) \rightarrow R(x, y)\big)\Big)$$

Example

Example : 4
Define

$$W(x): \quad x \text{ is a woman}$$

$$R(x): \quad x \text{ is rich}$$

$$R_1(x): \quad x \text{ is a real number}$$

$$R_2(x): \quad x \text{ is a rational number}$$

Using existential quantifier, symbolize the following statements.

(a) There exists a woman.

(b) Some women are rich.

(c) Some real numbers are rational numbers.

Algebraic Proof

(a)
Statement (a) can be restated as
"There exists an x such that x is a woman"

$\therefore$ Its symbolic form is

$$(\exists x)(W(x))$$

(b)
Statement (b) can be restated as
"There is an x, x is a woman and x is rich"

$\therefore$ Its symbolic form is

$$(\exists x)(W(x) \wedge R(x))$$

(c)
Statement (c) can be restated as
"There is an x, x is a real number and x is a rational number"

$\therefore$ Its symbolic form is

$$(\exists x)(R_1(x) \wedge R_2(x))$$

Example

Example : 5
Let

$$P(x): \quad x \text{ is a person}$$

$$T(x): \quad x \text{ trusts others}$$

$$R(x): \quad x \text{ is a rewarded}$$

$$G(x): \quad x \text{ is good}$$

$$Q(x): \quad x \text{ is teasing}$$

Symbolize the following statements:

(a) Some people who trust others are rewarded.

(b) If any one is good then John is good.

(c) Some one is teasing.

Algebraic Proof

Solution :
(a)
Statement (a) can be restated as

"There exists an x, x is a person, x trusts others and x is rewarded"

$\therefore$ Its symbolic form is $(\exists x)(P(x) \wedge T(x) \wedge R(x))$

(b)
Statement (b) can be restated as

"If there exists one x, x is a person and x is good, then John is good".

$\therefore$ Its symbolic form is $(\exists x)(P(x) \wedge G(x)) \rightarrow G(j)$

$$\left.\begin{array}{l} G(x): \ x \text{ is good} \\ G(j): \ \text{John is good} \end{array}\right\}$$

(c)
Statement (c) can be restated as

"There is one x, x is a person and x is teasing"

$\therefore$ Its symbolic form is $(\exists x)(P(x) \wedge Q(x))$

Example

Example : 6
Let

$$R(x): \quad x \text{ is a road}$$

and

$$D(x): \quad x \text{ lead to Denmark}$$

Symbolize the following statements:

It is not true that all roads lead to Denmark.

Algebraic Proof

Solution :
Consider "All roads lead to Denmark".
This can be rephrased as

"For all x, if x is a road then x leads to Denmark".

$\therefore$ Its symbolic form is

$$(\forall x)(R(x) \to D(x))$$

i.e.,

$$(\forall x)(\neg R(x) \vee D(x)) \qquad \{\because P \to Q \Leftrightarrow \neg P \vee Q\}$$

Now, the given statement can be rephrased as

"It is false that, for all x, if x is a road then x leads to Denmark".

$\therefore$ Its symbolic form is

$$\neg\Big((\forall x)(\neg R(x) \vee D(x))\Big)$$

i.e.,

$$\neg(\forall x)\neg(\neg R(x) \vee D(x))$$

i.e.,

$$(\exists x)(\neg\neg R(x) \wedge \neg D(x))$$

i.e.,

$$(\exists x)(R(x) \wedge \neg D(x))$$

Remark

If we want to negate a statement involving a quantifier then change the quantifier from universal to existential or from existential to universal and negate the statement which is quantified.

PREDICATE FORMULA

Definition

Predicate Formula :
The n–place predicate along with n individual variables is called an n–place predicate formula.
For example,

$$P(x_1, x_2, \ldots, x_n)$$

denotes an n–place predicate formula in which the letter P is an n–place predicate and

$$x_1, x_2, \ldots, x_n$$

are individual variables.

Example

Example :

$$P(x, y),\ R(x),\ Q(x, y, z)$$

are all predicate formulae.

Remark

- Predicate formulae are also known as atomic formulae.
- A 0–place predicate formula is a special case of predicate formula.

Definition

Free and Bound Variables :

1. The variable is said to be bound if it is concerned with either universal $(\forall x)$ or existential $(\exists x)$ quantifier.
2. The scope of the quantifier is the formulae immediately following the quantifier.
3. The variable which is not concerned with any quantifier is called free variable.

Example

Example :
Consider the following formulae:

1. $(\forall x)[P(x, y)]$

   In this formula, x is said to be bound and y is said to be free and the scope of the quantifier is $P(x, y)$.

2. $(\exists x)P(x) \wedge Q(x)$

   Here, x is bound in $(\exists x)P(x)$

   $$x \text{ is free in } Q(x)$$

   Scope of the quantifier $(\exists x)$ is $P(x)$.

3. $(\exists x)(P(x) \vee Q(x))$

   Here, x is bound in

   $$(\exists x)(P(x) \vee Q(x))$$

   Scope of the quantifier :

   $$P(x) \vee Q(x)$$

4. $(x)(P(x) \to Q(x))$

   Here, x is bound in

   $$(x)(P(x) \to Q(x))$$

   Scope of the quantifier is

   $$P(x) \to Q(x)$$

5. $(x)[P(x) \to (\exists y)R(x, y)]$

   Here, x is bound in

   $$(x)[P(x) \to (\exists y)R(x, y)]$$

   $$y \text{ is bound in } (\exists y)R(x, y)$$

   Scope of (x) is

   $$P(x) \to (\exists y)R(x, y)$$

   Scope of $(\exists y)$ is

THE UNIVERSE OF DISCOURSE :

Definition

Variables which are quantified stands for only those objects which are members of a particular set or class. Such a set is called the Universe of discourse or domain or simply universe.
The universe may be, the class of human beings, or numbers or some other objects.
The truth value of a statement depends upon the universe.

Example

Example : 1
Let

$$P(x) : x = x^2$$

be the statement.
If the universe of discourse is the set of integers, what are the truth values of

(a) $P(-1)$

(b) $(\forall x)\, P(x)$

Algebraic Proof

Solution :
Given:

$$P(x) : x = x^2$$

Universe of discourse

$$= \{0, \pm 1, \pm 2, \ldots\}$$

(a)
False. Since $x = -1$ does not satisfy $x = x^2$.

$$\therefore \quad \text{Truth value of } P(-1) \text{ is false.}$$

(b)
False. Since $x = 2$ does not satisfy $x = x^2$.

$$\therefore \quad \text{Truth value of } (\forall x)\, P(x) \text{ is false.}$$

Example

Example : 2
Let

$$P(x): \quad x \text{ is an even integer}$$

and

$$R(x,y): \quad x \text{ is divisible by } y$$

Let the universe of discourse be the set

$$U = \{1, 2, 4, 8, 16, 32\}$$

Find the truth values of the following:

$$(a)\ P(16) \qquad (b)\ P(4) \qquad (c)\ R(1,4) \qquad (d)\ R(16,2)$$

Algebraic Proof

Solution :
Given:

$$P(x): \quad x \text{ is even integer} \qquad \cdots(1)$$

$$R(x,y): \quad x \text{ is divisible by } y \qquad \cdots(2)$$

$$U = \{1, 2, 4, 8, 16, 32\}$$

(a)

$$P(16):$$

16 is an even integer which is true. {From (1)}

$$\therefore \quad P(16) \text{ is true.}$$

(b)

$$P(4):$$

4 is an even integer which is true. {From (1)}

$$\therefore \quad P(4) \text{ is also true.}$$

(c)

$$R(1,4):$$

1 is divisible by 4 which is false. {From (2)}

$$\therefore \quad R(1,4) \text{ is false.}$$

(d)

$$R(16,2):$$

16 is divisible by 2. {From (2)}

Example

Example : 3
Let

$$P(x): \quad x < 32$$

and

$$Q(x): \quad x \text{ is a multiple of } 10$$

with universe of discourse as all positive integers.
Find the truth value of

$$(\exists x)\,(P(x) \to Q(x))$$

Algebraic Proof

Solution :
Given:

$$P(x): \quad x < 32$$

$$Q(x): \quad x \text{ is a multiple of } 10$$

Universe of discourse:

$$U = \{1, 2, 3, 4, \ldots\}$$

To obtain the truth value of

$$(\exists x)\,(P(x) \to Q(x))$$

Now, 20 is an element in U.

$$\therefore$$

$$P(20): \quad 20 < 32 \text{ is true.}$$

$$Q(20): \quad 20 \text{ is a multiple of 10 which is true.}$$

$$\therefore \quad P(20) \to Q(20) \text{ is true.}$$

i.e.,
There exists atleast one

$$x = 20$$

such that

$$P(x) \to Q(x)$$

is true.
Hence

$$(\exists x)(P(x) \to Q(x))$$

Example

Example : 4
Use quantifiers to express the statement

"Every Computer Science student needs a course in Discrete Mathematics".

Algebraic Proof

Solution :
Let the universe of discourse be the set of all Computer Science students.
Let

$$P(x) : x \text{ needs a course in Discrete Mathematics.}$$

The given statement can be reframed as

"For all x, x needs a course in Discrete Mathematics"

$$\therefore \quad \text{Symbolic form : } (\forall x)\, P(x)$$

THE THEORY OF INFERENCE FOR PREDICATE CALCULUS :

The rules of equivalent formulae, tautological implications, Rule P, Rule T, Rule CP, indirect method of proof for statement calculus can also be used here.

Apart from this, we have some more rules to be followed, involving quantifiers. The rules of specification called US & ES are used to eliminate quantifiers, whereas the rules of generalization called UG & EG are to be used to prefix the correct quantifier.

Result

Universal Generalization : (UG)

$$A(y) \Rightarrow (x)\, A(x)$$

Provided x is not free in any given premises and if x is free in a prior step which resulted from use of ES, then no variables introduced by that use of ES appear free in $A(x)$.

Result

Existential Generalization : (EG)

$$A(y) \Rightarrow (\exists x)\, A(x)$$

Result

Universal Specification : (US)

$$(x)\, A(x) \Rightarrow A(y)$$

Result

Existential Specification : (ES)

$$(\exists x)\, A(x) \Rightarrow A(y)$$

Provided y is not free in any given premises and also not free in any prior step of the derivation.

Example

Example : 6
Verify the validity of the following argument:

Lions are dangerous animals. There are lions. Therefore there are dangerous animals.

Algebraic Proof

Solution :
First, let us translate the given argument into symbolic form:
Let

$$L(x): \quad x \text{ is a lion}$$

$$D(x): \quad x \text{ is a dangerous animal}$$

From the given problem, we have

$$\text{Premises as } (\forall x)(L(x) \to D(x)), \quad (\exists x)(L(x))$$

and the conclusion is

$$(\exists x)(D(x))$$

$$\therefore$$

Symbolic form of the given argument is

$$(\forall x)(L(x) \to D(x)), (\exists x)(L(x)) \Rightarrow (\exists x)(D(x))$$

Algebraic Proof

$\{1\}$	1) $(\exists x)(L(x))$	Rule P
$\{1\}$	2) $L(y)$	Rule ES
$\{3\}$	3) $(\forall x)(L(x) \to D(x))$	Rule P
$\{3\}$	4) $L(y) \to D(y)$	Rule US
$\{1,3\}$	5) $D(y)$	Rule T ($\because$ Modus Ponens) $(P,\ P \to Q \Rightarrow Q)$
$\{1,3\}$	6) $(\exists x)(D(x))$	Rule EG

Hence the argument is valid.

Example

Example : 7
Show that

$$(x)(H(x) \to M(x)) \wedge H(s) \Rightarrow M(s)$$

Algebraic Proof

Solution :

$\{1\}$	1) $(x)(H(x) \to M(x))$	Rule P
$\{1\}$	2) $H(s) \to M(s)$	Rule US
$\{3\}$	3) $H(s)$	Rule P
$\{1,3\}$	4) $M(s)$	Rule T $(P,\ P \to Q \Rightarrow Q)$

Remark

Here, whenever quantifier exist, we first remove the quantifier using either Rule US or Rule ES.

Example

Example : 8
Show that

$$(x)(P(x) \vee Q(x)) \Rightarrow (x)P(x) \vee (\exists x)Q(x)$$

Algebraic Proof

Solution :
We shall use the indirect method of proof.

Method – 1 : Method of Contradiction
Assume

$$\neg[(x)P(x) \vee (\exists x)Q(x)]$$

as an additional premise.

$\{1\}$	1) $\neg[(x)P(x) \vee (\exists x)Q(x)]$	Assumed Premises
$\{1\}$	2) $(\exists x)\neg P(x) \wedge (x)\neg Q(x)$	Rule T (Demorgan's law)
$\{1\}$	3) $(\exists x)\neg P(x)$	Rule T $(P \wedge Q \Rightarrow P)$
$\{1\}$	4) $(x)\neg Q(x)$	Rule T $(P \wedge Q \Rightarrow Q)$
$\{1\}$	5) $\neg P(y)$	Rule ES
$\{1\}$	6) $\neg Q(y)$	Rule US
$\{1\}$	7) $\neg P(y) \wedge \neg Q(y)$	Rule T $(P, Q \Rightarrow P \wedge Q)$
$\{1\}$	8) $\neg(P(y) \vee Q(y))$	Rule T (Demorgan's)
$\{9\}$	9) $(x)(P(x) \vee Q(x))$	Rule P
$\{9\}$	10) $P(y) \vee Q(y)$	Rule US
$\{1, 9\}$	11) $[P(y) \vee Q(y)] \wedge \neg[P(y) \vee Q(y)]$	Rule T $(P, Q \Rightarrow P \wedge Q)$

which is nothing but false value.
Therefore by method of contradiction, we have

$$(x)(P(x) \vee Q(x)) \Rightarrow (x)P(x) \vee (\exists x)Q(x)$$

Method – 2 : Method of Contrapositive

$\{1\}$	1) $\neg[(x)P(x) \vee (\exists x)Q(x)]$	Assumed Premises
$\{1\}$	2) $(\exists x)\neg P(x) \wedge (x)\neg Q(x)$	Rule T (Demorgan's)
$\{1\}$	3) $(\exists x)\neg P(x)$	Rule T $(P \wedge Q \Rightarrow P)$
$\{1\}$	4) $(x)\neg Q(x)$	Rule T $(P \wedge Q \Rightarrow Q)$
$\{1\}$	5) $\neg P(y)$	Rule ES
$\{1\}$	6) $\neg Q(y)$	Rule US
$\{1\}$	7) $\neg P(y) \wedge \neg Q(y)$	Rule T $(P, Q \Rightarrow P \wedge Q)$
$\{1\}$	8) $\neg[P(y) \vee Q(y)]$	Rule T (Demorgan's)
$\{1\}$	9) $(\exists x)\neg[P(x) \vee Q(x)]$	Rule EG
$\{1\}$	10) $\neg[(x)(P(x) \vee Q(x))]$	Rule T (Apply $\neg$)

Hence, we have

$$\neg[(x)P(x) \vee (\exists x)Q(x)] \Rightarrow \neg[(x)(P(x) \vee Q(x))]$$

Example

Example : 9
Test the validity of the following argument:

If an integer is divisible by 10 then it is divisible by 2. If an integer is divisible by 2, then it is divisible by 3.

$\therefore$

The integer divisible by 10 is also divisible by 3.

Algebraic Proof

Solution :
First, let us translate the given argument into symbolic form:
Let

$$D_{10}(x): \quad x \text{ is divisible by } 10$$

$$D_2(x): \quad x \text{ is divisible by } 2$$

$$D_3(x): \quad x \text{ is divisible by } 3$$

From the given problem, we have
Premises as

$$(\forall x)(D_{10}(x) \to D_2(x)),\ (\forall x)(D_2(x) \to D_3(x))$$

and conclusion is

$$(\forall x)(D_{10}(x) \to D_3(x))$$

$\therefore$

Symbolic form of the given argument is

$$(\forall x)(D_{10}(x) \to D_2(x)),\ (\forall x)(D_2(x) \to D_3(x))$$

$$\Rightarrow (\forall x)(D_{10}(x) \to D_3(x))$$

$\{1\}$	1) $(\forall x)(D_{10}(x) \to D_2(x))$	Rule P
$\{1\}$	2) $D_{10}(y) \to D_2(y)$	Rule US
$\{3\}$	3) $(\forall x)(D_2(x) \to D_3(x))$	Rule P
$\{1,3\}$	4) $D_2(y) \to D_3(y)$	Rule US
$\{1,3\}$	5) $D_{10}(y) \to D_3(y)$	Rule T by Hypothetical Syllogism ($\because P \to Q,\ Q \to R \Rightarrow P \to R$)
$\{1,3\}$	6) $(\forall x)(D_{10}(x) \to D_3(x))$	Rule UG

$\therefore$